AtliQ

SQL Business Intelligence Analysis

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The Challenge

The Problem:
AtliQ is scaling, but decision velocity and margin discipline are not keeping pace with growth, which slows down pricing, product, and sales decisions and increases the risk of margin leakage and missed growth opportunities.

The Task:

- ✓ Translate 10 ad-hoc Business questions into SQL queries
- ✓ Validate outputs and define the key KPIs consistently
- ✓ Turn data into actionable insights
- ✓ Present recommendations to top management

4 Management Actions to Improve Margin and Forecasting

Executive Summary: Four Critical Insights

Unbalanced Growth

Unique products grew **36.33%** (245 → 334). Growth concentrated in **Accessories (+34)**; **Networking (+3)** lags.
Action: Rebalance Portfolio Investment
KPI: segment revenue share, margin by segment

Priority: High

Over-Discounting

All top 5 India customers get 29-31% discounts
Action: Discount guardrails + tiering.
KPI: avg discount %, gross margin/customer, exceptions count.

Risk: High (Margin Leakage)

Market Gaps

- Present in 8 APAC markets. Whitespace assessment recommended to prioritise next markets.
- 5 major ones (Vietnam, Thailand, Malaysia, Singapore, Taiwan)

Priority: Medium

Severe Seasonality

Q1 = 33.72% of annual sales, Q3 = only 9.99% of annual volume. Not managing this proactively
Action: Q3 demand-smoothing + inventory plan
KPI: Q3 volume share, inventory turns, forecast accuracy.

Risk: High (Planning Volatility)

Insight #1: Portfolio is expanding fast, But growth is uneven

36.33%

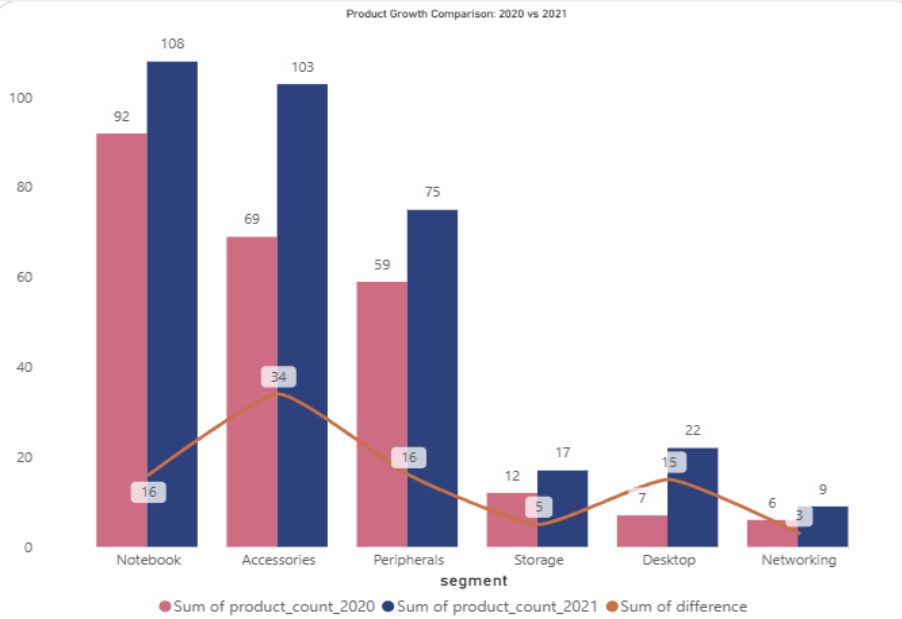
Product Growth YoY (245 - 334)

The Story:

✓ **Accessories: Product count grew** 49%, (69 -> 103, +34)

⚠ **Networking:** Only 2.3% of portfolio

Question: Are we investing in the segments that matter for revenue and margin?



💡 Recommendation: Rebalance Portfolio

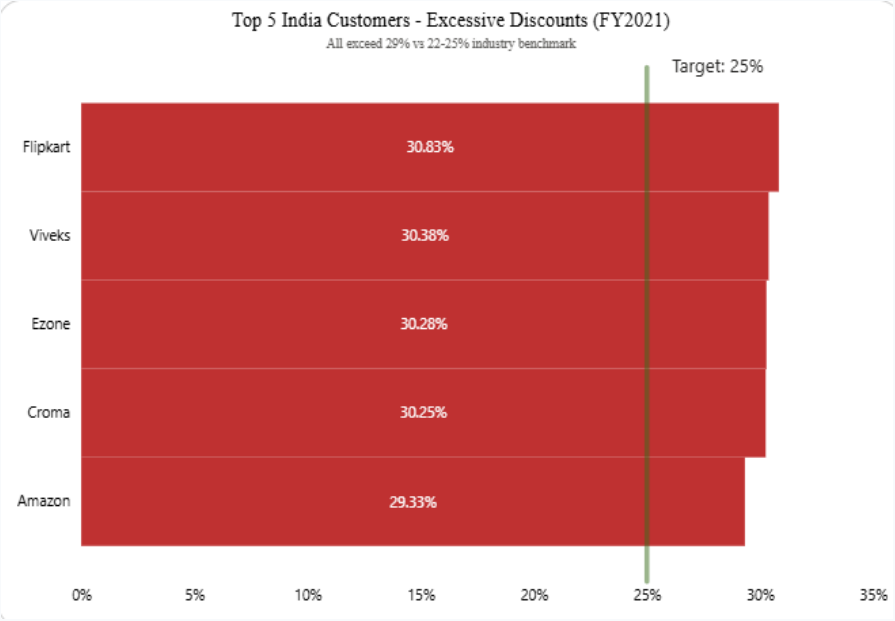
Tier Segment: Grow/ Maintain/ Rationalise
Focus: Under-represented, high-Margin Potential
Control: New-SKU approval thresholds

KPIs: Revenue Share, Margin, Sales/SKU

Insight #2: We're Leaving Money on the Table

€1.08M

Estimated annual margin leakage due to Annual Over-Discounting



The Story:

- All 5 top India customers: **29-31%**
- Industry benchmark: **22-25%**
- Gap: **5-9 percentage points**

💡 Recommendation: Tiered Discount Framework

Set Discount bands by account tier + Volume + Payment terms
Require approval for exceptions (Workflow)

KPI: Avg discount %, Margin by Customer and exception count

Insight #3: We're Missing Huge Opportunities

✓ Present In (8 Markets)

- IN India
- AU Australia
- JP Japan
- KR South Korea
- ID Indonesia
- PH Philippines
- BD Bangladesh
- NZ New Zealand

✗ Whitespace Assessment

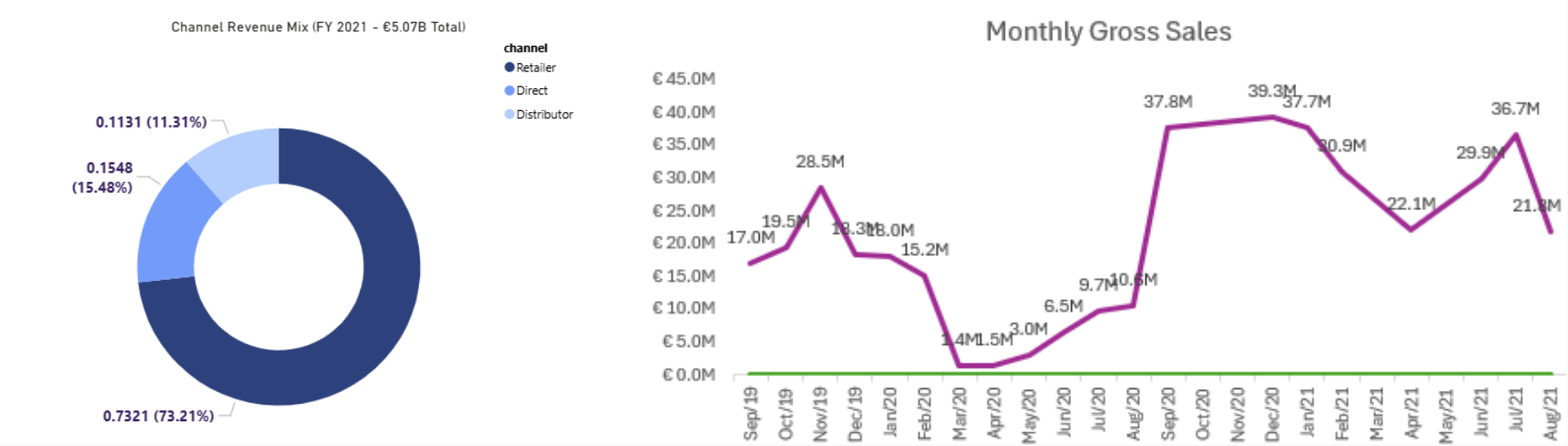
- Identify candidate markets using TAM, Channel readiness and Feasibility
- Score and shortlist Top 3 Markets for FY planning
- Validate with sales/channel stakeholders
- E.g. Vietnam, Thailand, Malaysia, Singapore

Recommendation: Run a 2 – week market prioritization sprint

Build a scorecard: TAM, Competition, pricing, channel, ops feasibility
Deliver: Top 3 market, Entry hypothesis, KPI targets

KPIs: pipeline value, CAC/Payback, quarterly revenue ramp

Insight #4: We Can't See Our Own Patterns



💡 **Recommendation:** Seasonality playbook + demand smoothing

- **Seasonality playbook + channel de-risking**

Plan inventory/marketing around Q1 peak; create Q3 demand-smoothing offers
Grow Direct channel in low-demand periods (B2B bundles, **targeted campaigns)**
KPIs: Q3 volume share, forecast accuracy, inventory turns, Direct sales share

Roadmap

1	**Portfolio Rebalancing** Rebalance growth across segments (reduce concentration)	Low Effective	High Impact
2	**Discount Framework** 3-tier system based on data	Invest: Minimal	High
3	**Market prioritisation (whitespace scorecard)**	Medium (with Investment)	Medium High
4	**Seasonal Strategy** Q1 Max Effort + Q3 Resilience	Medium	High

The Transformation

❌ BEFORE

- Manual Excel-based analysis for recurring questions
- Insights take days and are not consistent
- Limited visibility into trends across products, discounts, and seasonality
- Decisions rely on intuition rather than metrics

✓ AFTER

- SQL outputs standardised and repeatable
- Management-ready dashboard for rapid review
- Patterns visible at a glance (growth, discounting, seasonality, channel mix)
- Decisions supported by agreed KPIs

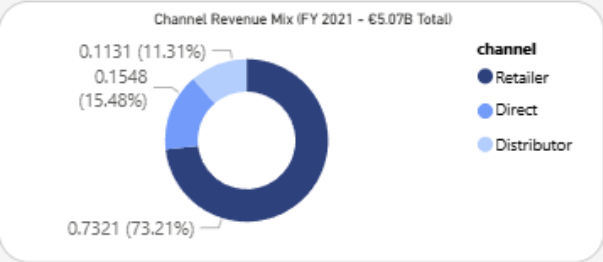
Not Just Queries

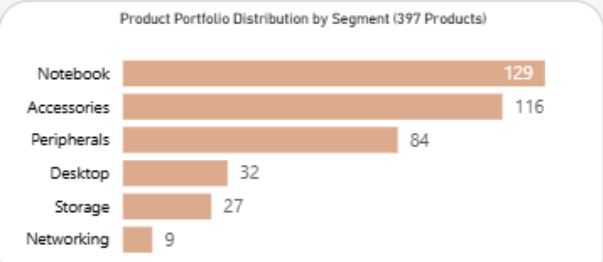
A DECISION ENGINE

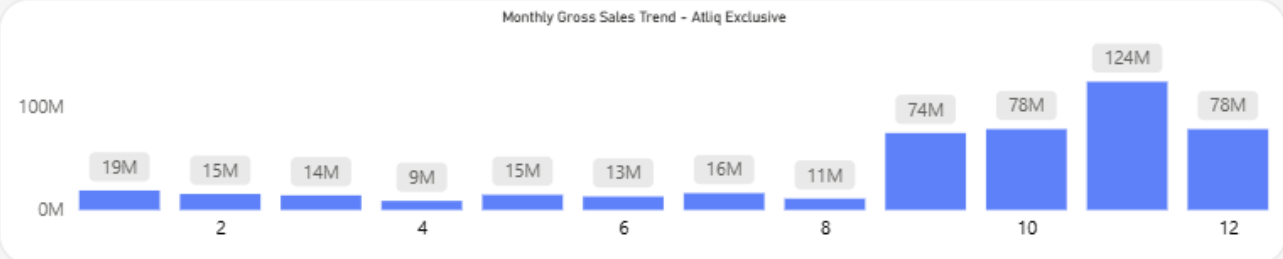
AtliQ Hardware Ad_Hoc Analysis (FY 2020 - 2021)

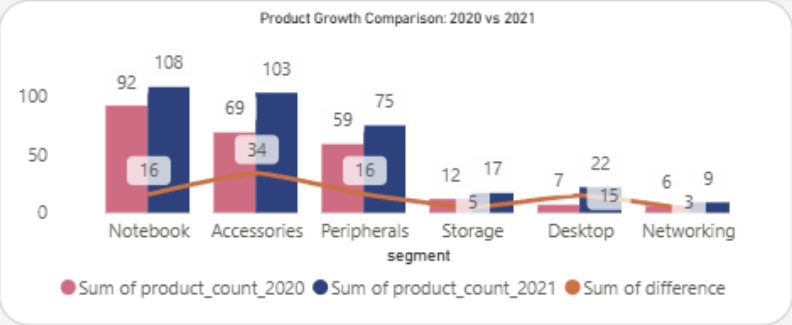
Unique Products (2020)	Unique Products (2021)	% change
245	334	36.33%

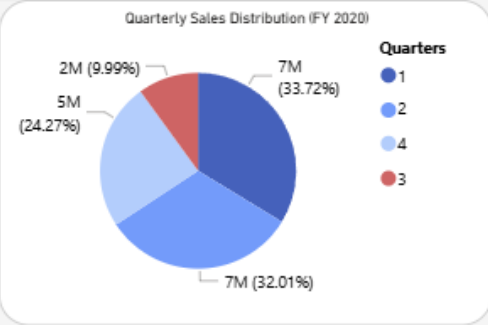
Highest Production Cost	Lowest Production Cost
$264.2861	$0.8654





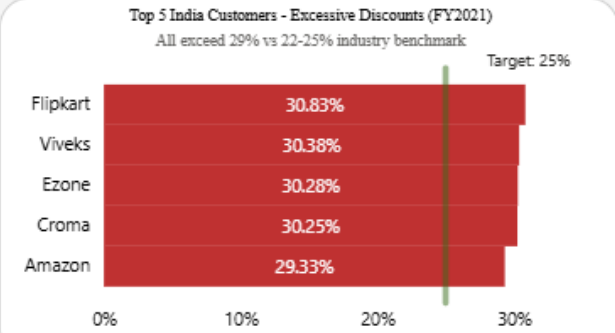






Top_3_Products_by_devision

division	product	product_code	Sum of Rank_Order	Sum of Total_sold_quantity
N & S	AQ Pen Drive 2 IN 1	A6720160103	1	701373
N & S	AQ Pen Drive DRC	A6818160202	2	688003
N & S	AQ Pen Drive DRC	A6819160203	3	676245
P & A	AQ Gamers Ms	A2319150302	1	428498
P & A	AQ Maxima Ms	A2520150501	2	419865



Thank You

From Data to Decisions

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https://www.linkedin.com/in/harshil-patel-188b2274/
https://github.com/HP85-NL/SQL-Portfolio-AtliQ-Hardware

Full SQL Code & Analysis on GitHub